

QUBE Motor Pre-Lab

1

Consider the following rotational mechanical system with inertia J_1 and J_2 connected by shafts with rotational stiffnesses K_1 and K_2 , respectively, with inputs $T_1(t)$ and $T_2(t)$ and outputs ω_1 and ω_2 . Assume the rods are rigid.

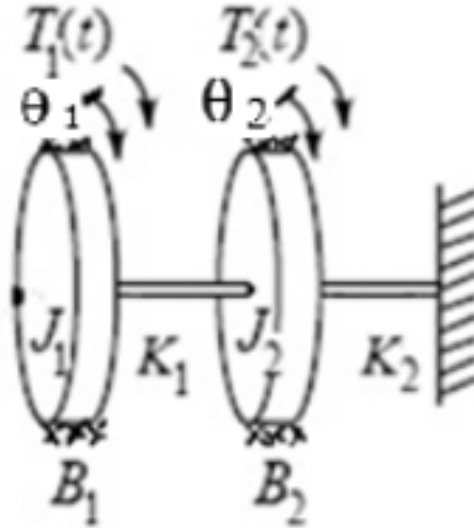


Figure 1: Schematic for rotational mechanical system for Q1”.

1.a

Use the rotational form of N2L on disks 1 and 2 to get:

$$\sum \tau_1 = J_1 \ddot{\theta}_1 = T_1 - B_1 \dot{\theta}_1 - K_1(\theta_1 - \theta_2) \quad (1)$$

$$\sum \tau_2 = J_2 \ddot{\theta}_2 = T_2 - B_2 \dot{\theta}_2 - K_2 x_2 - K_1(\theta_2 - \theta_1) \quad (2)$$

With Equation 1 and Equation 2, we can express this system with two differential equations:

$$J_1 \ddot{\theta}_1 + B_1 \dot{\theta}_1 + K_1(\theta_1 - \theta_2) = T_1 \quad (3)$$

$$J_2 \ddot{\theta}_2 + B_2 \dot{\theta}_2 + K_2 x_2 + K_1(\theta_2 - \theta_1) = T_2 \quad (4)$$

1.b

First, let our state vector and its derivative be:

$$x = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \quad \dot{x} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \quad (5)$$

Note that:

$$\dot{\theta}_1 = [0 \ 0 \ 1 \ 0] \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + [0 \ 0] \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}, \quad \dot{\theta}_2 = [0 \ 0 \ 0 \ 1] \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + [0 \ 0] \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$$

From Equation 1 and Equation 2, we can get equations to express $\ddot{\theta}_1$ and $\ddot{\theta}_2$ as a linear combination of the elements of x :

$$\ddot{\theta}_1 = J_1^{-1} (T_1 - B_1 \dot{\theta}_1 - K_1 (\theta_1 - \theta_2)) = - \left(\frac{K_1}{J_1} \right) \theta_1 + \left(\frac{K_2}{J_1} \right) \theta_2 - \left(\frac{B_1}{J_1} \right) \dot{\theta}_1 + \frac{T_1}{J_1} \quad (6)$$

$$= \begin{bmatrix} -\frac{K_1}{J_1} & \frac{K_2}{J_1} & -\frac{B_1}{J_1} & 0 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{J_1} & 0 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$$

$$\ddot{\theta}_2 = J_2^{-1} (T_2 - B_2 \dot{\theta}_2 - K_2 (\theta_2 - \theta_1)) \quad (7)$$

$$= \left(\frac{K_1}{J_2} \right) \theta_1 - \left(\frac{K_1 + K_2}{J_2} \right) \theta_2 - \left(\frac{B_2}{J_2} \right) \dot{\theta}_2 + \frac{T_2}{J_2}$$

$$= \begin{bmatrix} \frac{K_1}{J_2} & -\left(\frac{K_1 + K_2}{J_2} \right) & 0 & -\frac{B_2}{J_2} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{J_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$$

Therefore, the state-space representation of this system is:

$$\dot{x} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{K_1}{J_1} & \frac{K_2}{J_1} & -\frac{B_1}{J_1} & 0 \\ \frac{K_1}{J_2} & -\left(\frac{K_1 + K_2}{J_2} \right) & 0 & -\frac{B_2}{J_2} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{J_1} & 0 \\ 0 & \frac{1}{J_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} \quad (8)$$

$$y = \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} \quad (9)$$

2

Consider the system portrayed in Figure 2.

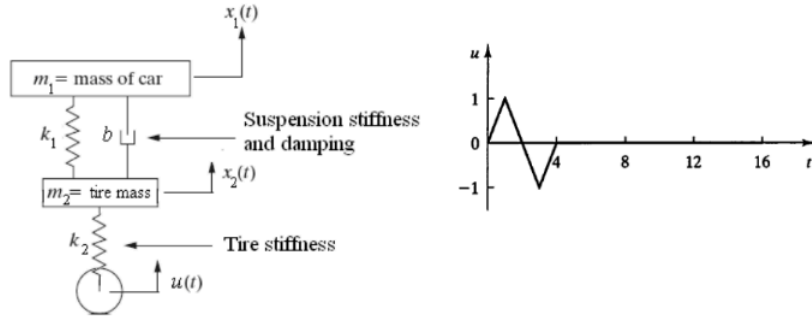


Figure 2: Schematic for mechanical system for Q2".

2.a

Let the state space vector be:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}, \quad \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} \quad (10)$$

Recall from ESC251 HW2, we derived the following differential equations for this corresponding system (for a reminder for how we derived this, see the Appendix in Section 3):

$$m_1 \ddot{x}_1 + b(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_2) = 0 \quad (11)$$

$$m_2 \ddot{x}_2 + b(\dot{x}_2 - \dot{x}_1) + k_1(x_2 - x_1) + k_2 x_2 = k_2 u \quad (12)$$

Note that:

$$\dot{x}_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} [u], \quad \dot{x}_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} [u]$$

From Equation 11 and Equation 12, we can get equations to express $\ddot{x}_1$ and $\ddot{x}_2$ as a linear combination of the elements of x :

$$\begin{aligned} \ddot{x}_1 &= m_1^{-1}(-b(\dot{x}_1 - \dot{x}_2) - k_1(x_1 - x_2)) \\ &= -\left(\frac{k_1}{m_1}\right)x_1 + \left(\frac{k_2}{m_2}\right)x_2 - \left(\frac{b}{m_1}\right)\dot{x}_1 + \left(\frac{b}{m_1}\right)\dot{x}_2 \\ &= \begin{bmatrix} -\frac{k_1}{m_1} & \frac{k_2}{m_1} & -\frac{b}{m_1} & \frac{b}{m_1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} [u] \end{aligned} \quad (13)$$

$$\begin{aligned} \ddot{x}_2 &= m_2^{-1}(b(\dot{x}_2 - \dot{x}_1) - k_1(x_2 - x_1) - k_2(x_2 - u)) \\ &= \left(\frac{k_1}{m_2}\right)x_1 - \left(\frac{k_1 + k_2}{m_2}\right)x_2 + \left(\frac{b}{m_2}\right)\dot{x}_1 - \left(\frac{b}{m_2}\right)\dot{x}_2 + \left(\frac{k_2}{m_2}\right)u \end{aligned} \quad (14)$$

$$= \begin{bmatrix} \frac{k_1}{m_2} & -\left(\frac{k_1+k_2}{m_2}\right) & \frac{b}{m_2} & -\frac{b}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} \frac{k_2}{m_2} \end{bmatrix} [u]$$

Therefore, the state-space representation of this system is:

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{k_1}{m_1} & \frac{k_1}{m_1} & -\frac{b}{m_1} & \frac{b}{m_1} \\ \frac{k_1}{m_2} & -\left(\frac{k_1+k_2}{m_2}\right) & \frac{b}{m_2} & -\frac{b}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_2}{m_2} \end{bmatrix} [u] \quad (15)$$

$$y = [x_1] = [1 \ 0 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + [0][u] \quad (16)$$

2.b

Consider the following given values:

$$m_1 = (1000)\text{kg}, \quad b = (4000)\frac{\text{kg}}{\text{s}}, \quad k_1 = (20000)\frac{\text{N}}{\text{m}}, \quad k_2 = \left[(5000)\frac{\text{N}}{\text{m}}, (10000)\frac{\text{N}}{\text{m}} \right], \quad t = (16)\text{s}$$

Plugging these values into Equation 15 and Equation 16, and running these numbers through the Appendix Code in Section 4, yields the following figure:

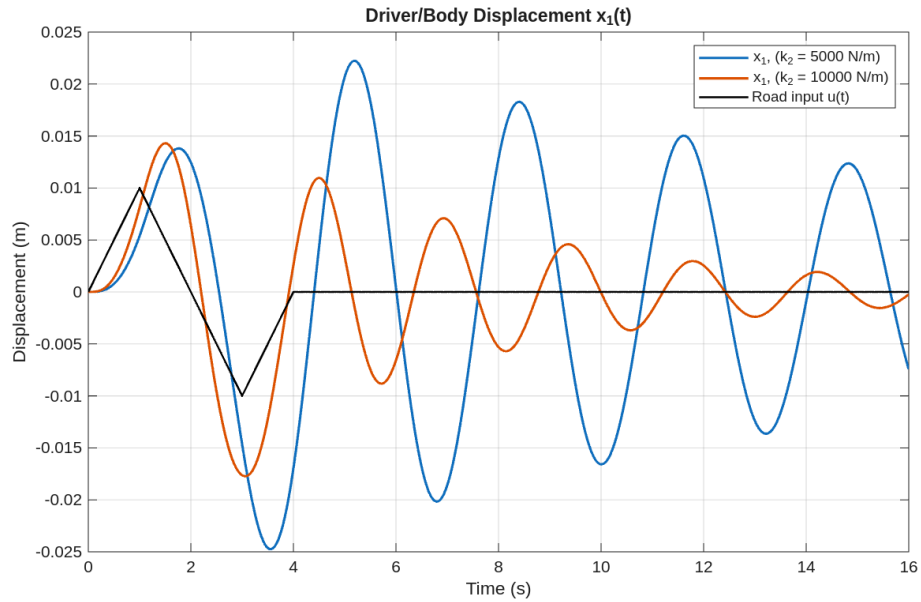


Figure 3: Driver/Body Displacement $x_2(t)$

2.c

As per Figure 3,

$$k_2 = (10000)\frac{\text{N}}{\text{m}}$$

resulted in a lower driver displacement.

3 Appendix

Refer to Figure 2 for the system corresponding to question 2.

By N2L for mass m_1 :

$$\sum F_y = m_1 \ddot{x}_1 = -b(\dot{x}_1 - \dot{x}_2) - k_1(x_1 - x_2) \quad (17)$$

By N2L for mass m_2 :

$$\sum F_y = m_2 \ddot{x}_2 = -b(\dot{x}_2 - \dot{x}_1) - k_1(x_2 - x_1) - k_2(x_2 - u) \quad (18)$$

These equations can be rewritten as Equation 11 and Equation 12 respectively.

4 Appendix Code

The following Matlab script was used to generate Figure 3.

```
m1 = 1000; % kg
m2 = 100; % kg
b = 4000; % kg/s
k1 = 20000; % N/m
k2_values = [5000 10000]; % N/m
t = 0:0.01:16;

% Triangle bump: 0 -> 1 cm -> -1 cm -> 0 over first 4 s
u1 = 0:0.01:1;
u2 = 0.99:-0.01:-1;
u3 = -0.99:0.01:0;
u4 = zeros(size(4.01:0.01:16));
u_cm = [u1 u2 u3 u4]; % bump in cm
u = 0.01 * u_cm; % convert to meters

% Output both displacements: y = [x1; x2]
C = [1 0 0 0];
D = [0];

y_store = cell(length(k2_values),1);

for i = 1:length(k2_values)
    k2 = k2_values(i);

    A = [0 0 1 0;
         0 0 0 1;
         -k1/m1 k1/m1 -b/m1 b/m1;
         k1/m2 -(k1+k2)/m2 b/m2 -b/m2];

    B = [0;
         0;
         0;
         k2/m2];

    sys = ss(A,B,C,D);
    y = lsim(sys,u,t);

    y_store{i} = y;
end

% Plot x1 (driver/body displacement)
figure;
plot(t, y_store{1}(:,1), 'LineWidth', 1.5); hold on;
plot(t, y_store{2}(:,1), 'LineWidth', 1.5);
plot(t, u, '--k', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Displacement (m)');
title('Driver/Body Displacement x_1(t)');
legend('x_1, (k_2 = 5000 N/m)', 'x_1, (k_2 = 10000 N/m)', 'Road input u(t)');
```